BHARGAVA SURYADEVARA

bhargava.suryadevara@gmail.com | (408) 210-4289 | Irving, TX | LinkedIn | GitHub | Portfolio

Professional Summary

AI/ML Engineer with 2+ years of experience designing, developing, and deploying machine learning systems and enterprise AI applications. Skilled in building agentic AI workflows, retrieval-augmented generation (RAG) systems, and scalable ML inference services, with hands-on experience in LLM orchestration, model evaluation, MLOps, cloud infrastructure, and enterprise automation.

Skills

Programming & Machine Learning: Python, SQL, Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow/Keras, XGBoost, Hugging Face Transformers, FastAPI, REST APIs

Generative AI & Agentic Systems: LangChain, LangGraph, Retrieval-Augmented Generation (RAG), LLM Agents, Model Context Protocol (MCP), Tool Calling, Prompt Engineering, Embeddings, Semantic Search, Vector Databases, Structured Outputs

MLOps & Cloud: MLflow, Docker, Kubernetes, CI/CD, GitHub Actions, AWS (S3, EC2, SageMaker), Microsoft Azure, Microsoft AI Foundry

Enterprise AI & Data Engineering: Microsoft Copilot Studio, Power Automate, Dataverse, SharePoint, Neo4j, PostgreSQL, Apache Spark

Professional Experience

AI Developer Intern, *Riddell* 05/2026 – Present | Irving, TX

- Designed and developed SignalGuide, an enterprise AI agent in Microsoft Copilot Studio, to evaluate business ideas against a 9-product software portfolio using Microsoft/SAP documentation, with structured recommendations and auditable Dataverse records.
- Engineered a hierarchical AI classification pipeline for 8,000 historical IT Helpdesk tickets, achieving 76% accuracy through iterative prompt optimization, error analysis, and confusion-pair evaluation.
- Developed an AI-powered self-service IT support agent integrating Microsoft Learn MCP, historical ticket knowledge, screenshot analysis, Power Automate, and Dataverse to support context-aware troubleshooting and automated escalation.
- Prototyped enterprise knowledge-graph solutions using Neo4j, Cypher, PostgreSQL, and Apache AGE to model SAP process, requirement, and document relationships, enabling graph-based knowledge retrieval through Copilot Studio REST tools.

AI/ML Engineer, *Augur Talentcare Pvt Ltd* 06/2022 – 05/2024 | Remote, India

- Designed and developed scalable ML inference APIs using FastAPI, Scikit-learn, TensorFlow/Keras, and PyTorch, enabling real-time model predictions and seamless integration with downstream applications.
- Implemented standardized MLOps workflows using Docker, Git, CI/CD, and MLflow for experiment tracking, model versioning, and deployment, reducing environment-related deployment issues by 32%.
- Processed and validated 500K+ records using Python, SQL, Pandas, NumPy, and Apache Spark, supporting data preprocessing, feature engineering, model evaluation, and batch inference pipelines.
- Automated AWS model deployment workflows using S3, EC2, SageMaker, Bash, and Docker, reducing deployment setup time from 3 hours to 45 minutes and improving production release efficiency.

Projects

Event Correlation Using Agentic AI in SOC [GitHub]

- Developed a LangGraph-based agentic AI pipeline to normalize and correlate security alerts using host, user, IP, event type, and temporal context, consolidating related signals into incident-level insights.
- Integrated RAG, embeddings, and semantic search to retrieve relevant security evidence and generate structured, context-aware incident summaries for SOC analyst investigations.
- Implemented structured LLM outputs, validation, and response-grounding mechanisms to minimize unsupported responses and improve reliability during security incident triage.

Hybrid Recommender System [GitHub]

- Developed a hybrid recommendation system combining collaborative filtering, user-item interactions, behavioral features, and item popularity to generate personalized rankings.
- Implemented data preprocessing, feature engineering, and PyTorch-based ranking workflows using Python, Pandas, NumPy, and Scikit-learn to address sparse user-item interactions.
- Achieved a **67% improvement in NDCG@20** over the baseline model, demonstrating significantly higher recommendation ranking quality.

Education

M.S. in Artificial Intelligence, San José State University | GPA: 3.736/4.0 08/2024 – 05/2026

B.Tech in Computer Science & Engineering, Vellore Institute of Technology | GPA: 3.78/4.0 08/2020 – 05/2024